

The Characteristic Equation

An invariant beneath the representations of a linear recurrence

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Abstract

A linear recurrence with constant coefficients can be presented as a row operator on a triangular array, as a companion matrix on a state vector, as a continuant determinant, as a generating function, and as a continued fraction. We treat these as coordinate systems on a single object rather than as competing accounts. The object is the characteristic equation. We make the convergence explicit through worked examples, exhibit the bidiagonal and tridiagonal matrices that generate the standard combinatorial triangles, and identify the transformation group under which the characteristic equation is the invariant. The matrix presentations are fixed only up to a diagonal gauge that preserves a certain product while leaving its factors free; the symmetry class of a representative, real symmetric, Hermitian, or complex symmetric, is therefore a choice within the gauge rather than a property of the sequence. Interactive companions to this article generate the triangles and their spines directly from their matrix data.

1 A generating triangle as a matrix action

Fix an infinite lower-triangular array $T(n, k)$, with $n \geq 0$ the row index and $0 \leq k \leq n$ the column index. Read each row as a vector $v_n = (T(n, 0), T(n, 1), \dots)^\top$. A two-term recurrence of the form

$$T(n, k) = \alpha(n, k) T(n-1, k) + \beta(n, k) T(n-1, k-1)$$

is exactly the statement that $v_n = L_n v_{n-1}$, where L_n is the lower bidiagonal matrix

$$L_n = \begin{pmatrix} \alpha(n, 0) & & & \\ \beta(n, 1) & \alpha(n, 1) & & \\ & \beta(n, 2) & \alpha(n, 2) & \\ & & \ddots & \ddots \end{pmatrix},$$

carrying α on the main diagonal and β on the subdiagonal. The array is the orbit of the seed vector v_0 under the operators $L_1, L_2, \dots$, and the entire content of the triangle is contained in the two functions α, β together with the seed.

This presentation makes the standard triangles into points in a small parameter space. The binomial triangle takes $\alpha = \beta = 1$ on the seed $e_0 = (1, 0, 0, \dots)$. The Eulerian triangle, whose entries encode Worpitzky's identity $x^n = \sum_k \langle n, k \rangle \binom{x+k}{n}$ [12, 3, 5, 2, 9, 7], takes $\alpha = k+1$ and $\beta = n-k$. The Stirling triangle of the second kind takes $\alpha = k$ and $\beta = 1$. The Delannoy array, genuinely two-dimensional [1], requires one further ingredient: the bidiagonal operator is post-composed with the row accumulator U , the cumulative-sum operator with $U_{kj} = 1$ for $j \leq k$, acting on the all-ones seed. The bidiagonal and Riordan-group organisation of such triangles is developed in [10, 11]. Table 1 records these.

The relationships among these triangles are now legible as relationships among their data. The binomial and Delannoy triangles share the operator L_n and differ only in the accumulator

Triangle	$\alpha(n, k)$	$\beta(n, k)$	Accumulator	Seed	Row-sum spine
Binomial (Pascal)	1	1	—	e_0	2^n
Eulerian / Worpitzky	$k + 1$	$n - k$	—	e_0	$n!$
Stirling (2nd kind)	k	1	—	e_0	Bell numbers
Delannoy	1	1	U	1	central Delannoy

Table 1: Four combinatorial triangles as points in the space of bidiagonal data $(\alpha, \beta, \text{accumulator}, \text{seed})$. The operator L_n has α on its diagonal and β on its subdiagonal; U is the row accumulator; $e_0 = (1, 0, 0, \dots)$ and **1** = $(1, 1, 1, \dots)$. The triangles are not isolated objects but neighbouring choices of weight.

and seed; the Delannoy array is the binomial triangle composed with cumulative summation on a different starting vector. The Stirling triangle differs from the binomial in a single weight, $\alpha = k$ in place of $\alpha = 1$. The Eulerian triangle alone has both weights varying with position. None of this is visible while the triangles are presented as completed diagrams; all of it is immediate once they are presented as matrix data.

Appendix A sets out each of these triangles explicitly, with its first rows displayed, its generating operator written in the form of the opening recurrence, and a single entry of each computed by hand from that recurrence, so that the reader may see the abstract formula instantiated in each case. A reader meeting these objects for the first time may wish to consult the appendix alongside this section.

1.1 Stirling as a weighted binomial recurrence

The Stirling case deserves emphasis, because the relationship it bears to the binomial triangle is usually left implicit. The binomial recurrence is

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1},$$

and the Stirling recurrence is

$$S(n, k) = k S(n-1, k) + S(n-1, k-1).$$

These are the same recurrence but for the weight on the first term: Stirling places the column index k where the binomial places 1. The binomial triangle is the constant-weight specialisation of Stirling at $k = 1$, or equally, Stirling is the binomial recurrence with a multiplier the binomial had silently set to unity.

The weighting propagates to the row sums. The binomial row sums are 2^n , the powers of the constant total weight. The Stirling row sums are the Bell numbers [8], and they satisfy

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k.$$

Here the binomial coefficients are the kernel of the recurrence. Each Bell number is the convolution of the entire preceding sequence against a row of the binomial triangle. The triangle that generates Stirling and the triangle that weights the recurrence for its row sums are the same triangle. Pascal appears inside the construction of its own near-relation, twice over: once as the

unweighted skeleton of the Stirling recurrence, and once as the convolution kernel for the Bell numbers.

The Bell recurrence has full depth; B_{n+1} depends on every earlier term. Its natural matrix presentation is therefore not bidiagonal but upper Hessenberg, with binomial entries filling the band above the diagonal. The bandwidth of the matrix records the depth of the recurrence, and the hierarchy from binomial through Stirling to Bell is a hierarchy of bandwidth.

2 One sequence, five matrices, one equation

Restrict now to a single spine and a constant-coefficient recurrence, where the structure is cleanest. The shallow diagonals of the binomial triangle sum to the Fibonacci numbers, and the projection of the binomial recurrence onto the shallow diagonal is the Fibonacci recurrence itself:

$$\sum_i \binom{n-i}{i} = \sum_i \left[\binom{n-1-i}{i} + \binom{n-2-(i-1)}{i-1} \right] = F_{n-1} + F_{n-2}.$$

Take this projected sequence, $F_n = F_{n-1} + F_{n-2}$, as the object of study, and ask each representation the same question: what does it say about the number whose powers solve the recurrence?

Iteration. Substituting a pure power $F_n = x^n$ and dividing by x^{n-2} ,

$$x^2 = x + 1.$$

Companion matrix. Writing the recurrence as $v_n = Av_{n-1}$ with

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad \det(A - \lambda I) = \lambda^2 - \lambda - 1.$$

The equation is now the characteristic polynomial of a matrix, which is where the term *characteristic equation* originates. Its roots are the eigenvalues that govern the growth; from $\det A = -1$ one obtains Cassini's identity $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$ with no induction, since $\det(A^n) = (\det A)^n$.

Continuant determinant. The tridiagonal matrix

$$M_n = \begin{pmatrix} 1 & 1 & & & \\ -1 & 1 & 1 & & \\ & -1 & 1 & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & -1 & 1 \end{pmatrix}$$

has, by cofactor expansion along the last row, $\det(M_n) = \det(M_{n-1}) + \det(M_{n-2})$, again the Fibonacci recurrence, again the same characteristic equation. Where the companion matrix is small and moves through time, the continuant is large and stands still; its determinant performs the whole iteration at once.

Generating function. The ordinary generating function

$$G(x) = \sum_{n \geq 0} F_n x^n = \frac{x}{1 - x - x^2}$$

carries $1 - x - x^2$ in its denominator, the reciprocal polynomial of $x^2 - x - 1$; the poles of G are the reciprocals of the roots.

Continued fraction. The golden ratio $[1; 1, 1, 1, \dots]$ is the fixed point of $x \mapsto 1 + 1/x$, so $x = 1 + 1/x$, hence $x^2 = x + 1$.

Five representations of differing character, analytic and algebraic, finite and infinite, static and dynamic, all forced to the same quadratic. The representations vary freely. The equation is invariant under the variation.

3 The gauge freedom and its group

The continuant of the previous section was not canonical. Its determinant depends on the off-diagonal entries only through their product. The matrix

$$M_n(b, c) = \begin{pmatrix} 1 & b & & \\ c & 1 & b & \\ & c & 1 & \ddots \\ & & \ddots & \ddots \end{pmatrix}, \quad \det(M_n) = \det(M_{n-1}) - bc \det(M_{n-2}),$$

reproduces Fibonacci for any pair with $bc = -1$. The choices

$$(b, c) = (1, -1), \quad (2, -\tfrac{1}{2}), \quad (i, i)$$

all satisfy $bc = -1$ and yield the identical determinant sequence. The freedom is a genuine gauge symmetry: conjugating by a diagonal matrix $D = \text{diag}(d_1, d_2, \dots)$ sends $b \mapsto (d_i/d_{i+1})b$ and $c \mapsto (d_{i+1}/d_i)c$, preserving the product bc and, with it, the determinant.

The symmetry class of the representative is therefore a choice within the gauge, not a feature of the sequence. Taking $b = c$ real gives a real symmetric matrix. Taking $b = \bar{c}$ gives a Hermitian matrix. Taking $b = c = i$ gives a complex symmetric matrix that is not Hermitian. All three present Fibonacci, because all three respect $bc = -1$.

To see that it is the invariant bc , and not the matrix silhouette, that determines the sequence, hold the shape and change the product. The choice $b = i, c = -i$ is Hermitian and has $bc = +1$. Its continuant satisfies

$$\det(M_n) = \det(M_{n-1}) - \det(M_{n-2}),$$

with characteristic equation $x^2 - x + 1 = 0$ and roots on the unit circle. The determinant sequence is the period-six pattern $1, 1, 0, -1, -1, 0, \dots$. The same matrix outline and the same magnitudes of entries, but a different product, give a different equation, and exponential growth gives way to oscillation. What separates Fibonacci from the period-six sequence is the value of the invariant bc , which is precisely the constant term of the characteristic equation.

This is the structure Klein's Erlangen Program [6] would recognise. A geometry, in Klein's organisation, is the study of the properties left invariant by a group of transformations. The transformations here are the diagonal conjugations $M \mapsto DMD^{-1}$, which permute the symmetry classes of the continuant, real symmetric, Hermitian, complex symmetric, while fixing the

determinant. The orbit of a continuant under this group is a family of matrices of every symmetry type sharing one determinant sequence. The invariant of the orbit is the characteristic equation. A sequence, on this reading, is not a matrix and not a recurrence but an equivalence class of presentations, and the characteristic equation is the label the gauge cannot alter.

3.1 Three actions, and what plays which role

The word “matrix” has now appeared in three distinct roles, and because all three involve diagonal objects and column vectors at some point, it is worth separating them explicitly. They are different operations and they do not interact.

The first is the *generative* action of Section 1. The bidiagonal operator L_n acts on a row vector v_{n-1} to produce the next row, $v_n = L_n v_{n-1}$. The vector is a row of actual triangle entries; the operator is the recurrence rule; the orbit of the seed v_0 is the triangle. This action steps forward in the row index n .

The second is the *companion* action of Section 2. For a recurrence of depth d , the state is the d -tuple of the most recent terms and the operator is the $d \times d$ companion matrix; for Fibonacci, $d = 2$ and the operator is the 2×2 matrix A . This action also steps forward, but the vector is now a small window of recent values rather than a row of the triangle, and the operator size is the depth of the recurrence.

The third is the *gauge* action of the present section. The diagonal matrix D does not act on any state vector. It acts on the continuant M itself, by conjugation $M \mapsto DMD^{-1}$, rewriting one operator as a similar operator with the same determinant. Nothing is being stepped forward; one is relabelling the basis of a static matrix.

Two cautions follow. First, the gauge matrix D and the seed vector v_0 are unrelated objects. The seed is the input to a generative iteration; the gauge is a change of basis applied to a fixed operator. They share nothing but the incidental presence of diagonal structure in their neighbourhood, and they never appear in the same equation.

Second, the depth of a recurrence surfaces differently in the two stepping pictures, and this is the right way to understand the otherwise puzzling difference in matrix shape between the companion matrix and the continuant. In the companion picture, depth is the *fixed dimension* of a square operator that is reused at every step: a depth- d recurrence has a $d \times d$ companion. In the continuant picture, depth is the *bandwidth* of a matrix whose size grows with the term index n : a depth- d recurrence gives a banded matrix with d active diagonals, and the matrix is $n \times n$ at the n th term. The companion pays the cost of depth once, in operator size, and reuses the operator; the continuant pays it continuously, as a band, in a matrix that grows. These are two accountings of the same quantity, which is why a tridiagonal continuant (bandwidth governed by depth two) and a 2×2 companion (dimension two) describe the same Fibonacci recurrence. The continuant is not acted upon as a state vector at all; it is evaluated, by its determinant. Its squareness records bandwidth and index, not a flattened state.

4 Why “hard to vary” is the right test

The multiplicity of representations might seem to threaten the idea that a sequence has a single best explanation. If iteration, companion matrix, continuant, generating function, and continued fraction all account equally for Fibonacci, and each can be exchanged for another at will, then by Deutsch’s criterion [4], that a good explanation is one that is hard to vary, the easy interchangeability looks like the signature of explanations whose details do no work.

The resolution is that the five are not five explanations but five coordinate systems on one explanation, and the one explanation is hard to vary exactly as the criterion requires. The

coefficients of the characteristic equation are not independently adjustable. Move the linear coefficient and Fibonacci becomes Pell, $x^2 = 2x + 1$; move the constant and it becomes Jacobsthal, $x^2 = x + 2$; raise the degree and one reaches Tribonacci, $x^3 = x^2 + x + 1$. In each case every representation moves together, because the eigenvalues, the closed-form bases, the generating-function poles, and the continued-fraction limit are all images of the same roots. No coefficient can be changed in one representation without changing it in all. That rigid coupling is what the phrase “hard to vary” names, and it is a property of the equation, not of any representation of it. Deutsch’s own observation that good explanations, being hard to vary, tend to be unique, applies here in a setting he was not addressing: there is one explanation of Fibonacci, the characteristic equation, and the abundance is an abundance of routes to it.

5 Interactive companions

Two browser tools accompany this article and make the matrix origins of the triangles manipulable directly. The first presents the bidiagonal operator L_n as an editable object: the user sets the weights $\alpha(n, k)$ and $\beta(n, k)$, toggles the row accumulator, chooses the seed, and watches the generated triangle and its spines respond, with the preset data of Table 1 available for the binomial, Eulerian, Stirling, and Delannoy triangles. The second tool generates the spines as continuant determinants, displaying the tridiagonal matrix whose determinant produces each named sequence and the symmetry class of the chosen representative. In both, the matrix is the primary object and the array or sequence is the derived one, which is the inversion this article argues for.

6 Coda

The progression has a single shape. A combinatorial triangle, usually met as a diagram, is the orbit of a bidiagonal operator; the operator’s data place it among its neighbours in a small parameter space. A spine of the triangle, usually met as a list, is generated by several operators of differing kinds, each of which exposes the same characteristic equation. The matrix representations of that equation form an orbit under a diagonal gauge group, and the equation is the invariant of the orbit. What is recognised first as an object, and then as a process, is finally recognised as an invariant preserved across the processes. The characteristic equation earns its name at the last stage: it is what is characteristic of the entire family of representations, the structure that survives translation among them.

A The triangles and their generating recurrences

Each triangle below is displayed in its first rows and then written as an instance of the opening recurrence

$$T(n, k) = \alpha(n, k) T(n - 1, k) + \beta(n, k) T(n - 1, k - 1), \quad (*)$$

with the convention that $T(n, k) = 0$ whenever $k < 0$ or $k > n$, and a stated seed for $T(0, 0)$ and, where needed, the rest of row zero. After each, one entry is computed by hand directly from $(*)$ so that the formula may be seen acting.

A.1 Binomial (Pascal), $\alpha = 1$, $\beta = 1$, seed e_0

$$\begin{array}{cccccccc}
& & & & 1 & & & \\
& & & & & 1 & & 1 \\
& & & 1 & & 2 & & 1 \\
& & 1 & & 3 & & 3 & & 1 \\
& 1 & & 4 & & 6 & & 4 & & 1 \\
& & 1 & & 5 & & 10 & & 10 & & 5 & & 1 \\
1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1
\end{array}$$

As an instance of (*): $\alpha(n, k) = 1$ and $\beta(n, k) = 1$, so

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

Worked entry. $\binom{4}{2} = \binom{3}{2} + \binom{3}{1} = 3 + 3 = 6$.

A.2 Eulerian / Worpitzky, $\alpha = k + 1$, $\beta = n - k$, seed e_0

The Eulerian number $\langle n, k \rangle$ counts permutations of $\{1, \dots, n\}$ with exactly k ascents. (The triangle below omits the trailing $\langle n, n \rangle = 0$.)

$$\begin{array}{ccccccc}
& & & & 1 & & \\
& & & & & 1 & & 1 \\
& & & 1 & & 4 & & 1 \\
& & 1 & & 11 & & 11 & & 1 \\
1 & & 26 & & 66 & & 26 & & 1 \\
\langle 6, k \rangle : & 1, & 57, & 302, & 302, & 57, & 1.
\end{array}$$

As an instance of (*): $\alpha(n, k) = k + 1$ and $\beta(n, k) = n - k$, so

$$\langle n, k \rangle = (k + 1) \langle n - 1, k \rangle + (n - k) \langle n - 1, k - 1 \rangle.$$

This recurrence is the engine of Worpitzky's identity $x^n = \sum_k \langle n, k \rangle \binom{x+k}{n}$, which expresses the monomial x^n in the basis of shifted binomial coefficients; the Eulerian triangle is thus the change-of-basis matrix from that basis to the monomials. The row sums are $n!$, since the ascents partition all $n!$ permutations. *Entry.* $\langle 4, 1 \rangle = (1 + 1) \langle 3, 1 \rangle + (4 - 1) \langle 3, 0 \rangle = 2 \cdot 4 + 3 \cdot 1 = 11$.

A.3 Stirling, second kind, $\alpha = k$, $\beta = 1$, seed e_0

The Stirling number $S(n, k)$ counts partitions of an n -set into exactly k non-empty blocks.

$$\begin{array}{cccccccc}
& & & & & 1 & & \\
& & & & & & 0 & & 1 \\
& & & & 0 & & 1 & & 1 \\
& & & 0 & & 1 & & 3 & & 1 \\
& & 0 & & 1 & & 7 & & 6 & & 1 \\
& 0 & & 1 & & 15 & & 25 & & 10 & & 1 \\
0 & & 1 & & 31 & & 90 & & 65 & & 15 & & 1
\end{array}$$

As an instance of (*): $\alpha(n, k) = k$ and $\beta(n, k) = 1$, so

$$S(n, k) = k S(n - 1, k) + S(n - 1, k - 1).$$

This is the binomial recurrence of A.1 with the constant weight 1 on the first term replaced by the column index k . The new element added to a partition either joins one of the k existing blocks, giving the $k S(n - 1, k)$ term, or forms a new block, giving the $S(n - 1, k - 1)$ term.

Entry. $S(4, 2) = 2 S(3, 2) + S(3, 1) = 2 \cdot 3 + 1 = 7$.

A.4 Bell numbers as a binomial convolution

The Bell number B_n is the total number of partitions of an n -set, so $B_n = \sum_k S(n, k)$ of A.3:

$$B_0, B_1, B_2, \dots = 1, 1, 2, 5, 15, 52, 203, 877, \dots$$

The Bell numbers do not satisfy a fixed-depth two-term recurrence and so are not the spine of a bidiagonal operator. They satisfy instead the full-depth binomial convolution

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k,$$

in which the kernel is a row of the binomial triangle A.1. This is the strong sense in which the Bell numbers are recursive-binomial: each term is the binomial-weighted sum of all previous terms. The natural matrix presentation is upper Hessenberg, with binomial entries above the diagonal. *Entry.* $B_4 = \binom{3}{0} B_0 + \binom{3}{1} B_1 + \binom{3}{2} B_2 + \binom{3}{3} B_3 = 1 + 3 + 6 + 5 = 15$.

A.5 Delannoy, $\alpha = 1$, $\beta = 1$, accumulator U , seed 1

The Delannoy number $D(n, k)$ counts lattice paths from $(0, 0)$ to (n, k) using steps east, north, and north-east. It is genuinely two-dimensional and is most naturally displayed as a square array rather than a triangle.

	0	1	2	3	4	5	6
0	1	1	1	1	1	1	1
1	1	3	5	7	9	11	13
2	1	5	13	25	41	61	85
3	1	7	25	63	129	231	377
4	1	9	41	129	321	681	1289
5	1	11	61	231	681	1683	3653
6	1	13	85	377	1289	3653	8989

The native recurrence is the three-term relation

$$D(n, k) = D(n - 1, k) + D(n, k - 1) + D(n - 1, k - 1),$$

whose extra term $D(n, k - 1)$ refers to the same row being filled. Writing L for the binomial bidiagonal operator of A.1 and U for the row accumulator $(Uw)_k = \sum_{j \leq k} w_j$, this same recurrence factors as $v_n = UL v_{n-1}$ on the all-ones seed $v_0 = \mathbf{1}$, which is the form recorded in Table 1: the Delannoy array is the binomial triangle post-composed with cumulative summation. The main diagonal $D(n, n)$ gives the central Delannoy numbers 1, 3, 13, 63, 321, 1683, 8989, ... *Entry.* $D(2, 2) = D(1, 2) + D(2, 1) + D(1, 1) = 5 + 5 + 3 = 13$.

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